Employee Payroll System

Output verived

Thank you so much for the time and effort taken to review my answer script

Agile Development Process and DevOps - Lab Fat

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Total Marks: 50

SET 1

ANSWER ALL THE QUESTIONS

Employee Payroll System

Problem Statement:
Ensure salaries are calculated and processed correctly.

Tasks:

1. Develop payroll calculation logic
2. Write JUnit tests for salary computation
3. Implement CI/CD pipeline with automated testing
    - Set up version control using GitHub
    - Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits
    - Automate build process using Maven
    - Integrate automated unit tests in pipeline
    - Configure CD pipeline for automatic deployment to staging/production
4. Deploy services on Docker and Kubernetes

- Containerize application using Docker (write Dockerfile)
- Build and push Docker image to container registry (Docker Hub)
- Create Kubernetes deployment and service YAML files

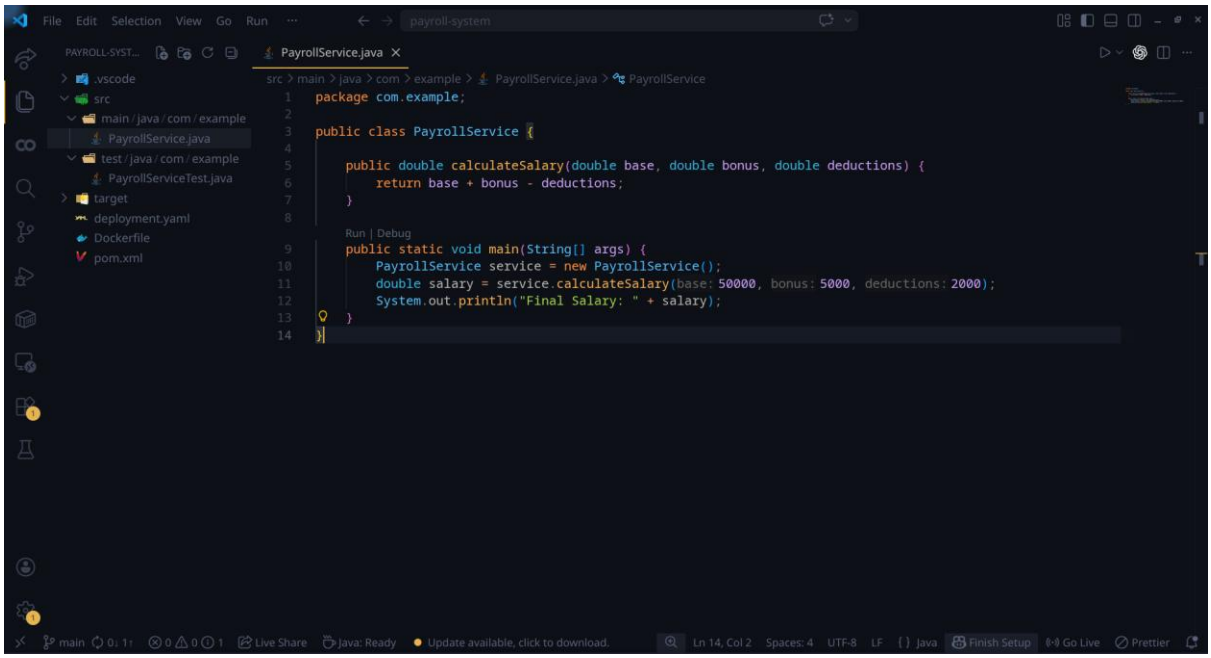
Components | Marks

- Java Program Development — 15
- JUnit Testing — 15
- CI/CD Pipeline Creation and Execution — 10
- Deploy services on Docker and Kubernetes — 10

Total: 50

ANSWER:

1. PayrollService.java



2. PayrollServiceTest.java

```java
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class PayrollServiceTest {

    @Test
    void testSalaryCalculation() {
        PayrollService service = new PayrollService();
        assertEquals(53000, service.calculateSalary(base: 50000, bonus: 5000, deductions: 2000));
    }
}
```

3. Output of Test Cases

```
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ payroll-system ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/muwafaq/dev/payroll-system/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ payroll-system ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/muwafaq/dev/payroll-system/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:3.0.0:test (default-test) @ payroll-system ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.PayrollServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.083 s - in com.example.PayrollServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.025 s
[INFO] Finished at: 2026-04-15T18:10:33+05:30
[INFO] ------------------------------------------------------------------------
```

4. Git Initialization Commands:

```
git init
git add .
```

git commit -m "Initial payroll system"

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

❯ git init
Initialized empty Git repository in /home/muwafaq/dev/payroll-system/.git/
❯ git add .
git commit -m "Payroll system with tests"
[main (root-commit) 2c33110] Payroll system with tests
 12 files changed, 126 insertions(+)
 create mode 100644 .vscode/settings.json
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/PayrollService.java
 create mode 100644 src/test/java/com/example/PayrollServiceTest.java
 create mode 100644 target/classes/com/example/PayrollService.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.PayrollServiceTest.xml
 create mode 100644 target/surefire-reports/com.example.PayrollServiceTest.txt
 create mode 100644 target/test-classes/com/example/PayrollServiceTest.class

~/dev/payroll-system  main
❯
```

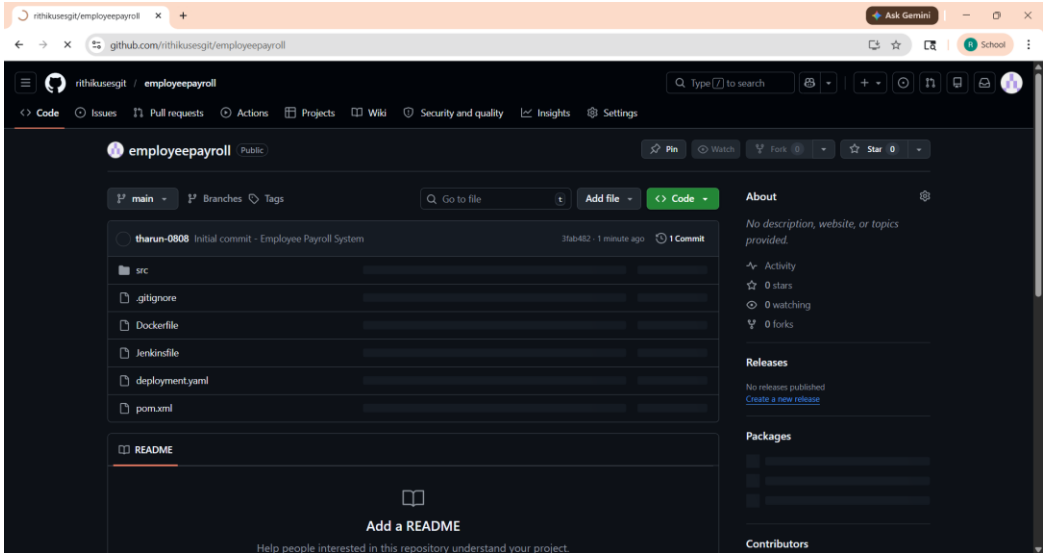
5. Push to GitHub

git remote add origin https://github.com/rithikusesgit/agilelabfat.git

git branch -M main

git push -u origin main

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Initialized empty Git repository in /home/muwafaq/dev/payroll-system/.git/
❯ git add .
git commit -m "Payroll system with tests"
[main (root-commit) 2c33110] Payroll system with tests
 12 files changed, 126 insertions(+)
 create mode 100644 .vscode/settings.json
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/PayrollService.java
 create mode 100644 src/test/java/com/example/PayrollServiceTest.java
 create mode 100644 target/classes/com/example/PayrollService.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.PayrollServiceTest.xml
 create mode 100644 target/surefire-reports/com.example.PayrollServiceTest.txt
 create mode 100644 target/test-classes/com/example/PayrollServiceTest.class
❯ git remote add origin https://github.com/rithikusesgit/agilelabfat.git
❯ git branch -M main
git push -u origin main
Enumerating objects: 38, done.
Counting objects: 100% (38/38), done.
Delta compression using up to 16 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (38/38), 4.89 KiB | 1.63 MiB/s, done.
Total 38 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/rithikusesgit/agilelabfat.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

~/dev/payroll-system  main
❯
```

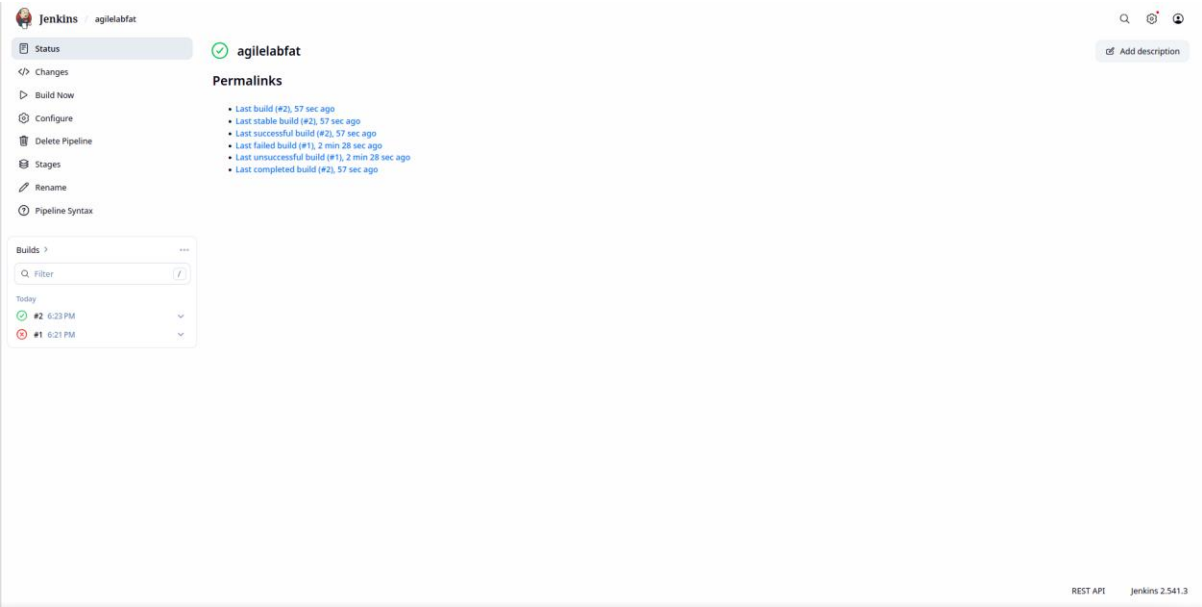


6. Jenkins CI Pipeline

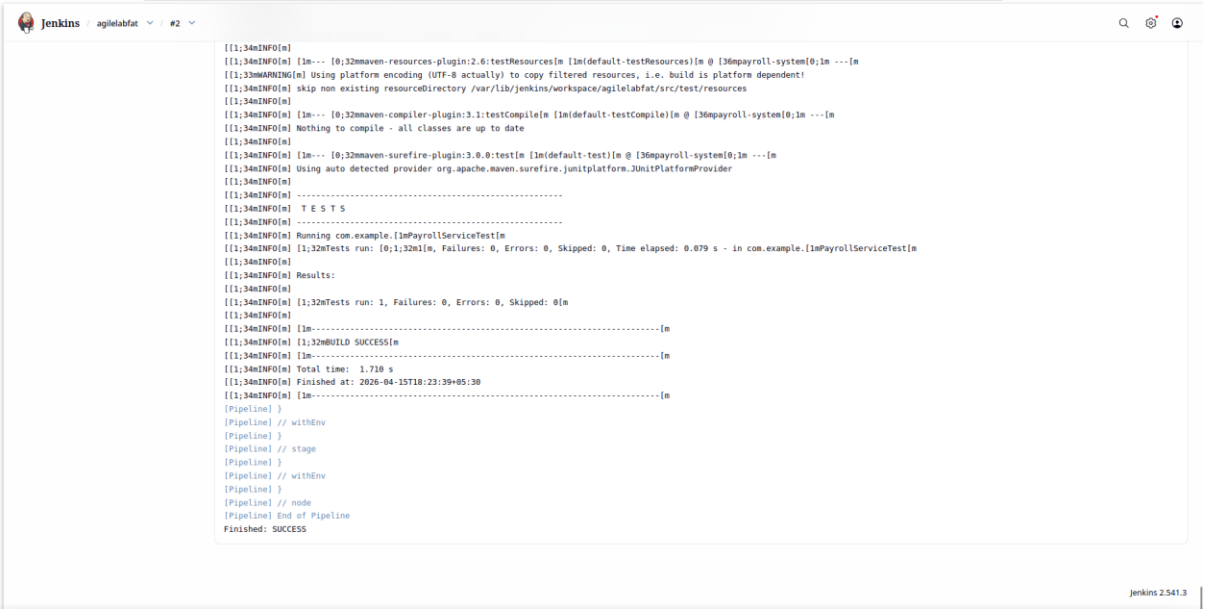
Pipeline Script

```
pipeline {
    agent any

    tools {
        maven 'Maven'
    }

    stages {

        stage('Clone') {
            steps {
                git branch: 'main', url:
'https://github.com/rithikusesgit/agilelabfat.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean install'
            }
        }
```

```
        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }
}
```



Build Success



7. Dockerfile

```
PayrollService.java        Dockerfile ×
Dockerfile > ...
1   FROM eclipse-temurin:17-jdk
2
3   WORKDIR /app
4
5   COPY target/*.jar app.jar
6
7   ENTRYPOINT ["java", "-jar", "app.jar"]
```

8. Build Docker Image

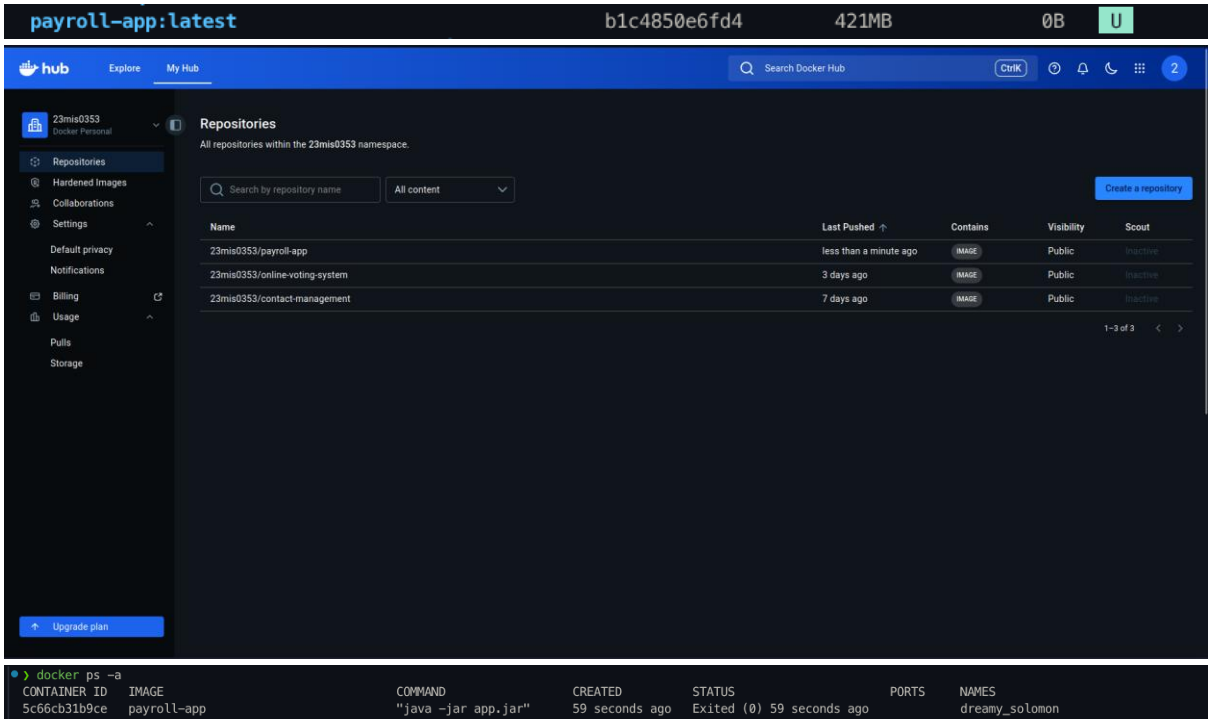
docker build -t payroll-app .

```
❯ docker build -t payroll-app .
[+] Building 2.1s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 145B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430
 => [internal] load build context
 => => transferring context: 2.77kB
 => CACHED [2/3] WORKDIR /app
 => [3/3] COPY target/*.jar app.jar
 => exporting to image
 => => exporting layers
 => => writing image sha256:b1c4850e6fd4955aa6924bee6d6e9ff214c7d4478d2224b4ba8816a57b8d1a3e
 => => naming to docker.io/library/payroll-app
❯ docker run payroll-app
Final Salary: 53000.0
```

9. Run Docker Container
docker run payroll-app

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ docker run payroll-app
Final Salary: 53000.0

~/dev/payroll-system  main
❯
```

10. Docker Images




```
❯ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                     PORTS     NAMES
5c66cb31b9ce   payroll-app    "java -jar app.jar"      59 seconds ago   Exited (0) 59 seconds ago            dreamy_solomon
```

11. Kubernetes Deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: payroll-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: payroll
  template:
    metadata:
      labels:
        app: payroll
    spec:
      containers:
        - name: payroll-container
          image: 23mis0353/payroll-app
```

12. Deploy to Kubernetes

kubectl apply -f deployment.yaml

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ kubectl apply -f deployment.yaml

deployment.apps/payroll-deployment unchanged

~/dev/payroll-system  main
❯
```

13. Verify Pods

kubectl get pods

```
❯ kubectl get pods
NAME                                    READY   STATUS             RESTARTS         AGE
client-c8b794fc-6tlt4                   1/1     Running            2 (5d18h ago)    13d
client-c8b794fc-pp25j                   1/1     Running            2 (5d18h ago)    13d
payroll-deployment-5659f4bbb6-9d527     0/1     Completed          8 (6m11s ago)    18m
server-795b5bbc8f-l2kxz                 0/1     CrashLoopBackOff   18 (3m20s ago)   16d
server-795b5bbc8f-xxz6s                 0/1     CrashLoopBackOff   19 (3m38s ago)   16d
```

14. View Logs

kubectl logs payroll-deployment-5659f4bbb6-9d527

```
❯ kubectl logs payroll-deployment-5659f4bbb6-9d527
Final Salary: 53000.0
```

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Thank you so much for the time and effort taken to review my answer script